

Workforce Bridge Program Pipeline Analysis

Pipeline attrition, employer quality, cost modeling, and fiscal-year budget reconstruction

An end-to-end people analytics investigation built from four and a half years of internship placement data at a multi-campus workforce development program. The analysis answers eight operational questions standard program reporting could not, using a synthetic dataset engineered to mirror the program's real structure, including its actual employer partnerships, role distributions, and compensation parameters.

1,223

PLACEMENTS ANALYZED, 5
CAMPUSES, 2022–2026

14.6%

OF COMPLETIONS
REACHED THE FULL 9-
MONTH TERM

 $r = 0.968$

QUALITY SCORE
CORRELATION WITH
COMPLETION RATE
($P < 0.001$)

\$1.51M

5-YEAR AVERAGE FISCAL
YEAR BUDGET REQUIRED
(INCL. COORDINATOR
TEAM)

KEY FINDINGS

Completion and full-term retention are different metrics. 55.2% of placements end in Planned Completion, but only 14.6% of those reach the full 9-month placement length. A distinction standard completion-rate reporting hides entirely.

Most attrition is early. 70.3% of student exits happen within 60 days of placement start, identifying the highest-leverage window for intervention.

Employer quality and cost are not the same thing. Cost per completed placement ranges from \$2,177 to \$17,461 across partners. Roughly an 8x spread invisible in completion-rate-only reporting.

The quality scoring model is statistically validated, not arbitrary. Confirmed via Pearson correlation testing against actual outcomes ($r = 0.968$, $p < 0.001$).

Campus completion rates differ by 15.5 percentage points, a gap large enough to justify investigating coordinator practice and intake process differences directly.

Fiscal-year budget figures reconcile exactly to the underlying wage data. Placements spanning fiscal-year boundaries are allocated proportionally by calendar day, verified to the cent.

METHODOLOGY

- Five-category outcome classification framework built from operational program knowledge
- Weighted employer quality scoring, statistically validated
- Cost-per-completion modeling using real compensation parameters
- Fiscal-year budget reconstruction with exact day-level wage allocation
- Early-exit risk prediction (Logistic Regression, Random Forest, Gradient Boosting compared via 5-fold CV)
- Concentration risk analysis (HHI-style index) of employer-role pairings

On the data: the dataset is synthetic, built to mirror the real structure of the program, including real employer partnerships, real role distributions, real campus volume patterns, and real compensation terms (\$18/hr, 17hr avg week, 9-month max placement). The generator is published in full alongside the analysis.

Built end-to-end: data generation, statistical analysis, predictive modeling, and policy recommendations. Every figure independently verified against source data before publication.